

Вариант 3;

Задание 1

1.1;

a = -1; b = 1;

f[x_] = 3^x;

n = 4;

Do[{Subscript[x, i] = Cos[Pi*(2*i + 1)/(2*(n + 1))], Subscript[f, i] = f[Subscript[x, i]]

P[x_] = Sum[Product[(x - Subscript[x, i])/(Subscript[x, k] - Subscript[x, i]), {i, 0, k -

$$\begin{aligned}
 & 1 - \frac{1}{4} \times 3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{5}(10-2\sqrt{5})} x + \frac{1}{4} \times 3^{1+\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{5}(10-2\sqrt{5})} x - \\
 & \frac{1}{4} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{10-2\sqrt{5}} x + \frac{1}{4} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{10-2\sqrt{5}} x + \\
 & \frac{1}{2} \times 3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{10}(5+\sqrt{5})} x - \frac{1}{2} \times 3^{1+\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{10}(5+\sqrt{5})} x - \\
 & \frac{1}{2} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{2}(5+\sqrt{5})} x + \frac{1}{2} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{2}(5+\sqrt{5})} x - 4x^2 + \\
 & 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2 + 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2 + 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2 + 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2 + \frac{3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2}{\sqrt{5}} + \\
 & \frac{3^{1+\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2}{\sqrt{5}} - \frac{3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2}{\sqrt{5}} - \frac{3^{1+\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2}{\sqrt{5}} + 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{5}(10-2\sqrt{5})} x^3 - \\
 & 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{5}(10-2\sqrt{5})} x^3 + \frac{1}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{10-2\sqrt{5}} x^3 - \\
 & \frac{1}{5} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{10-2\sqrt{5}} x^3 - 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{2}{5}(5+\sqrt{5})} x^3 + 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{2}{5}(5+\sqrt{5})} x^3 + \\
 & \frac{1}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{2(5+\sqrt{5})} x^3 - \frac{1}{5} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{2(5+\sqrt{5})} x^3 + \frac{16x^4}{5} - \\
 & \frac{4}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4 - \frac{4}{5} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4 - \frac{4}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4 - \frac{4}{5} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4 - \\
 & \frac{4 \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4}{\sqrt{5}} - \frac{4 \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4}{\sqrt{5}} + \frac{4 \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4}{\sqrt{5}} + \frac{4 \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4}{\sqrt{5}}
 \end{aligned}$$

```
Tbl = Table[{Subscript[x, i], Subscript[f, i]}, {i, 0, n}];
Pl[x_] = InterpolatingPolynomial[Tbl, x] // Simplify // Expand
Simplify[P[x] - Pl[x]] == 0
```

$$\begin{aligned}
& 3^{\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} - \frac{1}{2} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{10}(25-5\sqrt{5})} x + \frac{1}{2} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{10}(25-5\sqrt{5})} x - \\
& \frac{1}{2} \times 3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{10}(5-\sqrt{5})} x + \frac{1}{2} \times 3^{1+\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{1}{10}(5-\sqrt{5})} x + \\
& \frac{1}{2} \times 3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{10}(5+\sqrt{5})} x - \frac{1}{2} \times 3^{1+\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{10}(5+\sqrt{5})} x - \\
& \frac{1}{2} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{2}(5+\sqrt{5})} x + \frac{1}{2} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{1}{2}(5+\sqrt{5})} x + 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2 + \\
& 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2 - 4 \times 3^{\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2 + 3^{\frac{1}{2}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2 + \\
& 3^{\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} + \frac{1}{2} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2 + \frac{3^{1-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2}{\sqrt{5}} + \frac{3^{1+\frac{1}{2}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^2}{\sqrt{5}} - \\
& \frac{3^{1+\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} - \frac{1}{2} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2}{\sqrt{5}} - \frac{3^{1+\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} + \frac{1}{2} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^2}{\sqrt{5}} + \\
& \frac{1}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{2}{5}(25-5\sqrt{5})} x^3 - \frac{1}{5} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{2}{5}(25-5\sqrt{5})} x^3 + \\
& 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{2}{5}(5-\sqrt{5})} x^3 - 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} \sqrt{\frac{2}{5}(5-\sqrt{5})} x^3 - 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{2}{5}(5+\sqrt{5})} x^3 + \\
& 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{\frac{2}{5}(5+\sqrt{5})} x^3 + \frac{1}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{2(5+\sqrt{5})} x^3 - \frac{1}{5} \times 3^{\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} \sqrt{2(5+\sqrt{5})} x^3 - \\
& \frac{4}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4 - \frac{4}{5} \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4 + \frac{16}{5} \times 3^{\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4 - \\
& \frac{4}{5} \times 3^{\frac{1}{2}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4 - \frac{4}{5} \times 3^{\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} + \frac{1}{2} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4 - \frac{4 \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4}{\sqrt{5}} + \\
& \frac{4 \times 3^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4}{\sqrt{5}} - \frac{4 \times 3^{\frac{1}{2}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} x^4}{\sqrt{5}} + \frac{4 \times 3^{\frac{1}{4}} \sqrt{10-2\sqrt{5}}^{-\frac{1}{2}} \sqrt{\frac{1}{2}(5-\sqrt{5})} + \frac{1}{2} \sqrt{\frac{1}{2}(5+\sqrt{5})} x^4}{\sqrt{5}}
\end{aligned}$$

True

1.2;

```
M1 = Maximize[{D[f[x], {x, n + 1}], a <= x <= b}, x][[1]];
M2 = Minimize[{D[f[x], {x, n + 1}], a <= x <= b}, x][[1]];
M = Max[Abs[M1], Abs[M2]];
R = M/((n + 1)! * 2^n) // N
Print["Ответ: оценка погрешности R <= ", R]
```

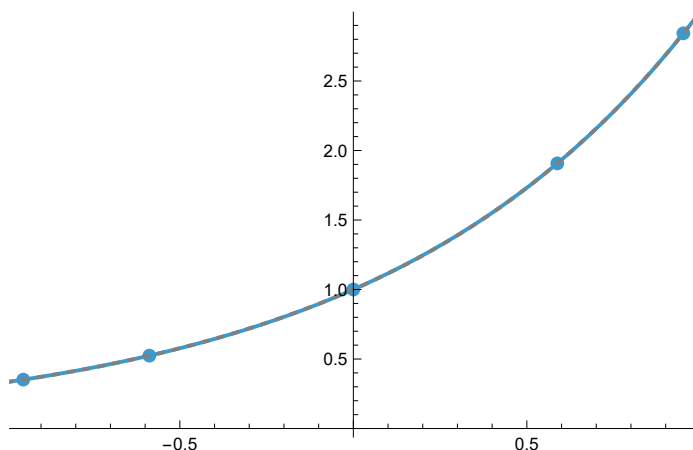
Ответ: оценка погрешности R <= 0.00250059

0.00250059

```

1.3;
Gr1 = ListPlot[Tbl, PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[f[x], {x, a, b}];
Gr3 = Plot[P[x], {x, a, b}, PlotStyle -> {Gray, Dashed}];
Show[Gr1, Gr2, Gr3]

```



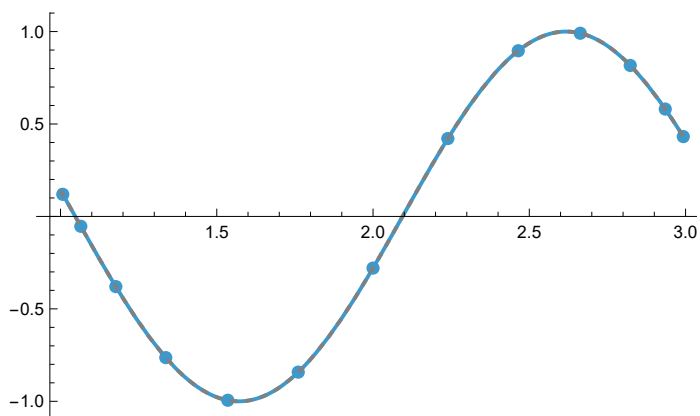
Задание 2

```

Clear[n, M, f, a, b, P, Pl];
f[x_] = Sin[3 x];
a = 1; b = 3; epsilon = 10^(-7);
M[n_] := Maximize[{Abs[D[f[x], {x, n + 1}]], a <= x <= b}, x][[1]];
n = 1;
While[N[(M[n] * (b - a)^(n + 1)) / (((n + 1)!) * 2^(2 n + 1))] > epsilon, n++];
Print["Требуемая степень многочлена n = ", n];
Do[{Subscript[x, i] = ((b - a)/2) * Cos[Pi * (2*i + 1) / (2*(n + 1))] + (a + b)/2, Subscript[f, i] = f[Subscript[x, i]]}, {i, 0, n}];
P[x_] = Sum[Product[(x - Subscript[x, k]) / (Subscript[x, k] - Subscript[x, i]), {k, 0, n, k != i}], {i, 0, n}];
Tbl = Table[{Subscript[x, i], Subscript[f, i]}, {i, 0, n}];
Gr1 = ListPlot[Tbl, PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[f[x], {x, a, b}];
Gr3 = Plot[P[x], {x, a, b}, PlotStyle -> {Gray, Dashed}];
Show[Gr1, Gr2, Gr3]

```

Требуемая степень многочлена n = 12



Задание 3

```

Clear[n, M, f, a, b, omega];
f[x_] = Sin[x];
a = 0; b = Pi/4; epsilon = 10^(-2);
Subscript[x, i_] := a + i*(b - a)/n;
n = 1;
While[
  (M = Maximize[{Abs[D[f[x], {x, n + 1}]], a <= x <= b}, x][[1]];
  omega[y_] = Product[y - Subscript[x, i], {i, 0, n}];
  Pomega = Maximize[{Abs[omega[y]], a <= y <= b}, y][[1]];
  N[M*Pomega/((n + 1)!)] > epsilon),
  n++];
Print["Требуемая степень n = ", n, "; минимальное число узлов = ", n + 1];
R = N[M*Pomega/((n + 1)!)]];
Print["Ответ: число равноотстоящих узлов ", n + 1, ", оценка R <= ", R]
Требуемая степень n = 8; минимальное число узлов = 9
Ответ: число равноотстоящих узлов 9, оценка R <= 0.00542992

```

Задание 4

```

Clear[n, M, f, a, b];
f[x_] = Cos[x];
a = Pi/2; b = 3 Pi/4; n = 2;
M = Maximize[{Abs[D[f[x], {x, n + 1}]], a <= x <= b}, x][[1]];
R = M*(b - a)^(n + 1)/(((n + 1)!)*2^(2 n + 1)) // N;
Print["Ответ: оценка погрешности по оптимальным узлам R <= ", R]
Ответ: оценка погрешности по оптимальным узлам R <= 0.0025233

```